

KEYS

1. Windows+ X = All system details
2. Alt + tab = Another window,
(If use shift,it can move forward and backward))
3. Windows R = run dialogue (command)
4. Windows+ H = voice to text in (microsoft word)
5. Windows+shift+s = snipping tool (edit screenshot while we take screenshot)
6. Windows +. = emoji (can use in word,notion,notepad)
7. ctrl + shift + esc = Task Manager
8. Alt + ctrl + dlt = when use in the system hanging time
9. Alt + f4 = minimize , alt + fn + f4 = all close
10. If we play screenrecord,its recording time less than 30 seconds,can make GIF
11. windows alt R = Instant record
12. Windows + G = Capture details+ record screen option
13. Cntrl shift arrow = select sentences, shift arrow also work
14. Cntrl down arrow = select nearest sentence also
15. Docs.new (search in chrome) = google new document
16. In documents:
 - 1.share criterias : Can restrict by view or not viewing for others
 - 2.clock icon,we can check changes(it can view)
 - 3.multiple person can access,give request - it will be get through him in mail
 - 4.@today - today's date
 - 5.@yesterday - yesterday's date
 6. file - new - template
 - 7.insert -symbol- eg: rupee symbol
 - 8 ctrl enter- next page
 - 9 file-page- edit page as portrait,page less, make titles-heading (alt ctrl 1),make sub headings
 - 10 if we need to paste a paragraph or screenshot,in the same format of doc- cntrl shift v
 11. ctrl k - link copy - site or pic anything
 - 12.ctrl forward slash- shortcuts
 - 13.ctrl alt f- can add foot note
 - 14.ctrl F - FIND any particular word
 - 15.ctrl H- FIND and REPLACE any word
 - 16 Tab- can move cursor instead of mouse (used in any website)
 - 17.Windows- accessibility - keyboard virtual
 18. word file can convert into docs by upload in on drive
17. **Notebooklm** (app) - if a pdf or file is large it can convert into short,specific details can ask
18. If need any pdf file use or search - filetype:pdf
19. Ctrl shift T = Redo close tab if we unexpectedly close

Data Science - Dr Ignez Semmelweiz

- **Data Science used in** Travel, Healthcare, Sales, Credit & Media, Marketing, Social Media, Automation
- **Automation-** self driving cars, Pilotless aircrafts,drones
- **Credit& Insurance** - Claims prediction, Fraud & Risk Detection
- **Marketing-** Upselling,cross selling,churn- churn meaning in business (leaving or stopping subscription)
- Data Science includes **AI, ML, Deep Learning, Maths, Statistics, Visualization, EDA**

Why ML for future - 460+ exabytes data generate every day based on 2025

- **1 byte= 8 bit**
- **1000 such letters form 1 kilobyte**

MACHINE LEARNING

1. Supervised learning - label image -train
- classification and regression

classification- predicting a category or class.

Eg: It is tiger or cat finding, email spam detection

Regression - predicting continuous numerical values

Eg: House price-400000, salary-20000, 45 degree temperature.

2. Unsupervised learning- no labels,machine will figure out some factors
-clustering and association includes here

Clustering-grouping similar data

Association- finding relationships or rules b/w items

😊 There is a site- "Teachable Machine" to train model

⇒ Machine Learning Applications-

- Virtual Assistant
- eCommerce
- Financial Services
- social media
- Health care

PYTHON

Python is a High level, object oriented, interpreter,Dynamic semantic, open source language

1.Numpy - is a python library for multi dimesional array processing- raw data handle(calculation,mathematical operation)

2.Pandas -Built on top of numpy,which provides multi dimesional array processing- data analysing,cleaning,data manipulation

3.scikit learn - machine library in python- (clustering,regression process)

AI

- Turing Test

♥ Process of DATA SCIENCE

- Data Mining- collecting relevant data
- Data Cleaning- handling missing data

- **Data Exploration- (EDA) Form hypothesis about defined problems (means observe data and making possible assumptions or predictions)**
- **Feature Engineering - eg: weekend, lot of people purchase a particular thing(finding through feature engineering (Changing or creating data columns in a better way)**
- **Predictive Modelling -training a model using old data to predict future results)**
- **Data Visualization**

To check python install?

Select terminal -> `python --version` OR `py -V`

If we get the version ,python is installed

Python style -snake case

JavaScript - Camel case

Virtual environment -venv(in built in python)

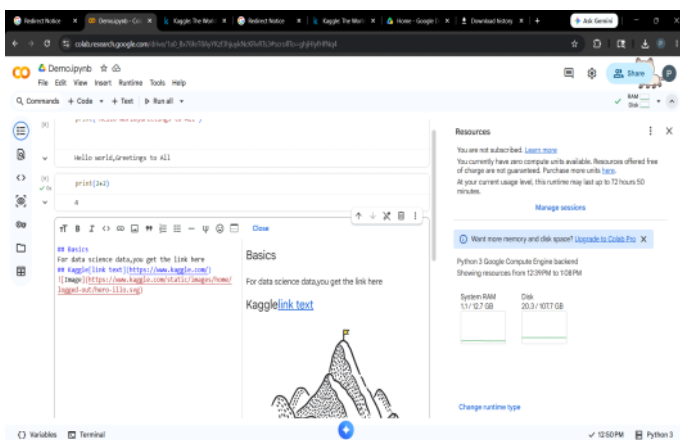
For Code demo use "[google colab.com](https://colab.research.google.com)" (chrome search)

Insert link is okay there

If we link any image copy address,image can see at right side by adding ! In left of image le,

Put `![Image]` then copy image address or link,we can see image at right side

Can see this in the screenshot 😊



In Google colab-can get any extra data through drive,

It is get by `files-> mount drive-` can get data

1. In colab,if we need command in code,Add a `#` symbol or `/` before the text
2. If I need output in multiple variables use bracket with comma `(4,5,6)` or `("red","yellow","blue")`
- 3.Reuse ,store->`variables`-it is a reserved memory location

Eg: `age = 12`

`print(age)`

Output= 12

4. Variables must be start with **letter or underscore, 0-9**

5. Variables can't be a keyword like **print**

6. **Case sensitive**

Modulus %- find reminder

Exponent *- find power

Floor division //-divide numbers, also remove the decimal part.eg: if 3.444,

OPERATORS

== Equal to

!= not equal

Relational operators

-

In set, we can't add duplicate elements,slicing,is only mutable,it is ordered

List functions

Reverse function

Iteration method- to avoid repeated print

-definite loop-for loop- one condition,1 increment,1 decrement

-indefinite loop-while loop:initialization, condition with relational operations,increment or decrement

More than conditions combine- using logical operations

User Defined Function in python

-built in fn

-

Control statements,mainly 3 statements in python

- Break,

- Continue,

- Pass-

Function call eg:mul()- inside the () is called the arguments

Actual parameter

Formal parameter

OOPS

Map fn-

Object and class

Inside the class, written fn- called->methods, variables called attributes

Inheritance-multiple objects create-parents class->inherit to child class

Multilevel inheritance

-class access-key called objects

My Lecture notes paatashala

22 May 2026 19:22

Correlation in ML- It is the Relationship between two variables

Causation means cause and effect relationship b/w two variables

But correlation is not causation

Version control Git & Github

02 June 2026 09:26

Version control repository- Local(PC/Laptop)

Remote control repository- Remote(Server)

1. Git status-git status is a Git command that shows the current state of your repository.
2. git init stands for **Git Initialize**.
It creates a new Git repository in your current folder.
3. git add
git commit -m "added file"
git add-second line add
git commit -m "added line1"
- 3 git log
- 4.git checkout
- 5.git switch
- 6 git branch
- 6.git checkout master
- 7.git branch
- 8.git branch dev
- 9.git checkout dev
- 10.git clone URL- code from github repository
11. Git push
12. Fork
13. Pull request